

PAPER_1652 — Pop III IMF Upper Bound = $M_{\max} = A_5 \times 2 = 120 M_{\odot}$ (top of Pop III IMF)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** First Stars **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (broader-corpus PAPER_13xx mine)

Observation

PAPER_1331 (PAPER_13xx broader corpus) gives:

$M_{\max} = A_5 \times 2 = 120 M_{\odot}$ (top of Pop III IMF)

Status: **EXACT**.

UQFF Closed Identity

$M_{\max} = A_5 \times 2 = 120 M_{\odot}$ (top of Pop III IMF) EXACT

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Cross-Domain Reuse of A_5

The icosahedral group order $A_5 = 60$ governs: - **Pop III IMF upper bound** = $A_5 \times 2 = 120 M_{\odot}$ (PAPER_1652) - **Quantum supremacy threshold** = $A_5 = 60$ qubits (PAPER_1655)

Vastly different physical scales: stellar masses ($\sim 10^2$ - 10^{33} kg) vs quantum qubits (countable integers). Same structural integer.

A_5 already appears in: - Heart rate = $A_5 + SO_5 = 70$ bpm (PAPER_1537) - SH0ES Hubble = $A_5 + SO_5 = 70$ km/s/Mpc (PAPER_1573) - Hayflick limit = $A_5 = 60$ (PAPER_1373) - T_SCm activation = $A_5 = 60$ K (PAPER_1505) - BH seed = $A_5 \times D_{BSFG}^2 \times D_{crit}$ (PAPER_1650)

$A_5 = 60$ is a **universal UQFF integer** spanning biology, cosmology, BH physics, quantum computing.

NOT REPLACEMENT

Empirical first stars measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1331
- Related: PAPER_1636-1645 (tier-22 broader-corpus closures); PAPER_1494-1635 (PAPER_1209 series)
- Calculator dispatch: `calculate_paradox({"paradox": "pop_iii_imf_max_120"})`

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